

Experiment No.14 SHIFT REGISTER	
Course Code: CPE 013	Program: BSCpE
Course Title: Logic Circuits and Design	Date Performed: 14/4/2026
Section: CPE22S2	Date Submitted: 14/4/2026
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1. Objective(s):	
This activity aims to implement and demonstrate a sequential logic circuit – shift register.	
2. Intended Learning Outcomes (ILOs):	
The students shall be able to: 2.1 Demonstrate and simulate the different mode of shift register. 2.2 Analyze and observe the process of each mode of shift register.	
3. Discussion:	
Flip-flop is a 1 bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop. Such a group of flip-flop is known as a Register. The n-bit register will consist of n number of flip-flop and it is capable of storing an n-bit word. The binary data in a register can be moved within the register from one flip-flop to another. The registers that allow such data transfers are called as shift registers. There are four modes of operations of a shift register. • Serial Input Serial Output (SISO) • Serial Input Parallel Output (SIPO) • Parallel Input Serial Output (PISO) • Parallel Input Parallel Output (PIPO)	

4. Resources:

- Personal Computer
 - Internet connection
- Simulation Tools:
- o Logically

5. Procedure:

1. Construct and simulate the block diagram of SISO as shown in Figure 14.1 in Logically.

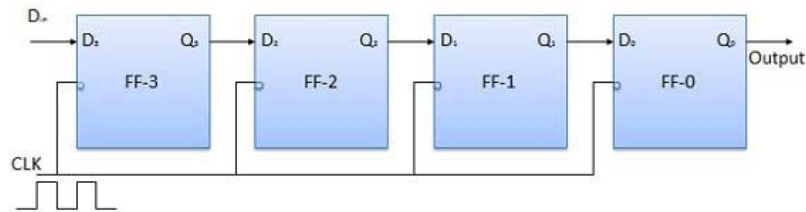


Figure 14.1. Serial Input Serial Output (SISO)

2. See the truth table for SISO in Table 14.1 and evaluate the result of your simulation. Screenshot the simulation result and place it in data and results in section 6.

	CLK	$D_3 = Q_2$	$Q_2 = D_2$	$Q_2 = D_1$	$Q_1 = D_0$	Q_0
Initially			0	0	0	0
(i)	↓	1	1	0	0	0
(ii)	↓	1	1	1	0	0
(iii)	↓	1	1	1	1	0
(iv)	↓	1	1	1	1	1

→ Direction of data travel

Table 14.1. Truth Table for SISO

3. Repeat step 1 for Figure 14.2, 14.3 and 14.4.

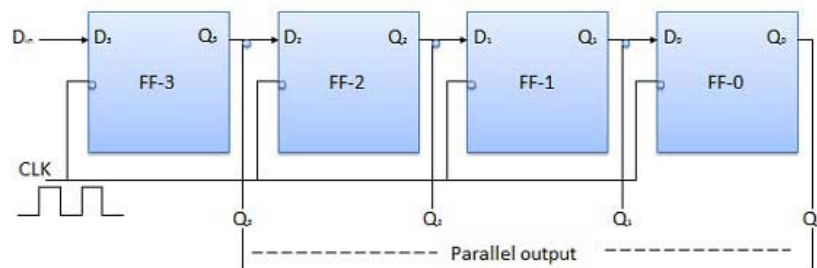


Figure 14.2. Serial Input Parallel Output (SIPO)

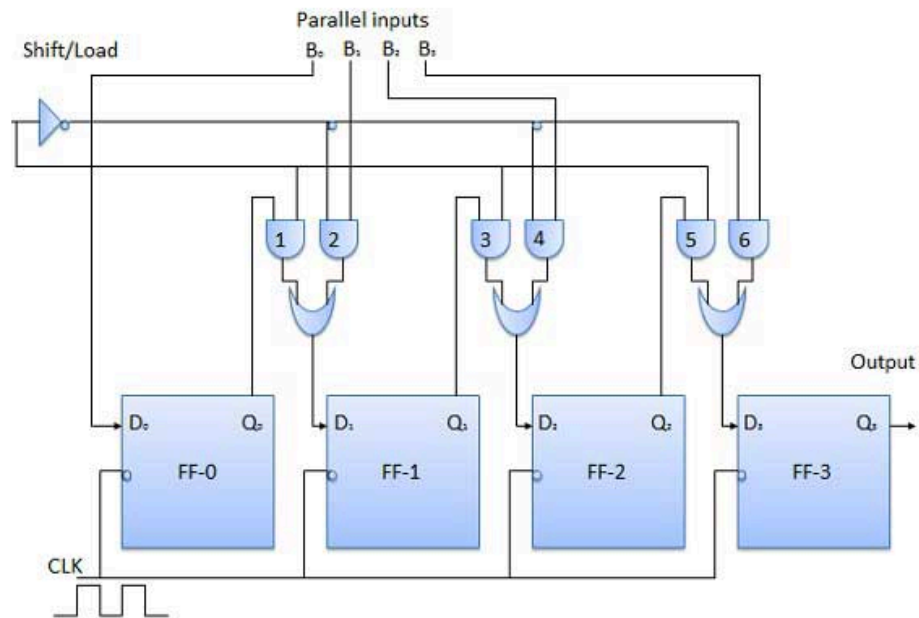


Figure 14.3. Parallel Input Serial Output (PISO)

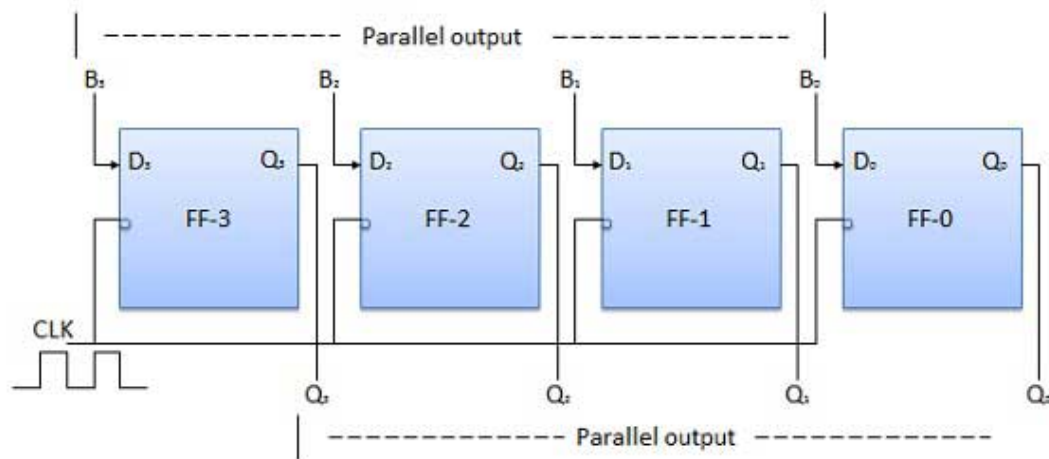


Figure 14.4. Parallel Input Parallel Output (PIPO)

4. Construct the truth table for SIPO, PISO and PIPO respectively for Table 14.2, Table 14.3 and Table 14.4 based on the result of your simulation in Step 3. Screenshot the simulation result and place it in data and results in section 6.

Table 14.2 - SIPO

Clock Pulse	Serial Input(SI)	Q0	Q1	Q2	Q3(parallel Ouput)
0	0	0	0	0	0000
1	1	1	0	0	1000
2	0	0	1	0	0100
3	1	1	0	1	1010
4	1	1	1	0	1101

Table 14.3 - PISO

Clock Pulse	Parallel Input(P3 P2 P1 P0)	Load/Shift	Serial Output(S0)
0	1011	Load	-
1	1011	Shift	1
2	1011	Shift	0
3	1011	Shift	1
4	1011	Shift	1

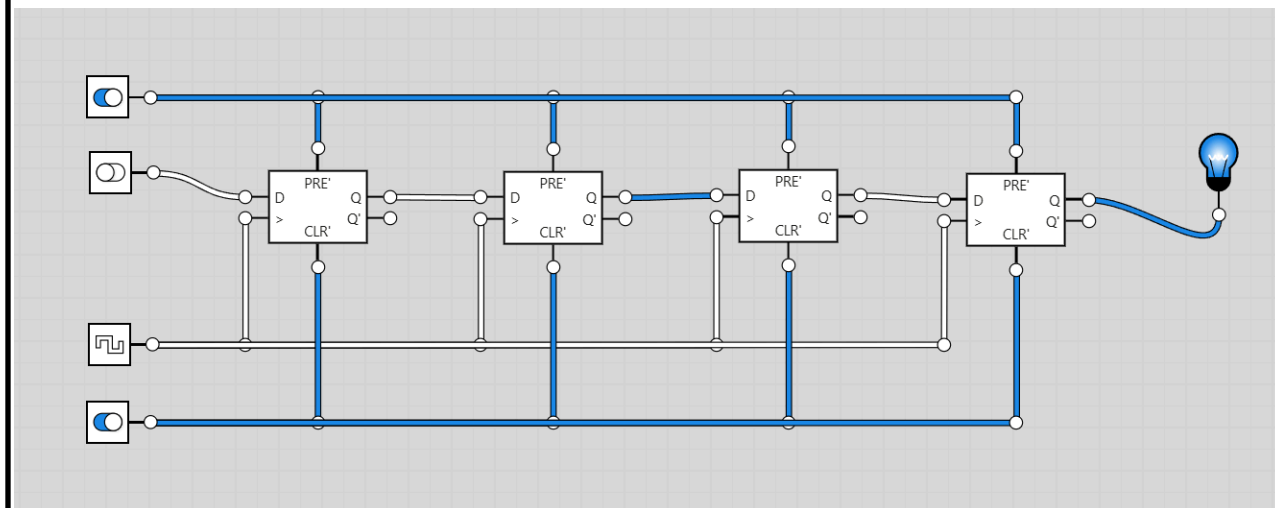
Table 14.4 - PIPO

Clock Pulse	Parallel Input(P3 P2 P1 P0)	Q3	Q2	Q1	Q0
0	1101	1	1	0	1
1	0110	0	1	1	0
2	1001	1	0	0	1
3	1110	1	1	1	0
4	0011	0	0	1	1

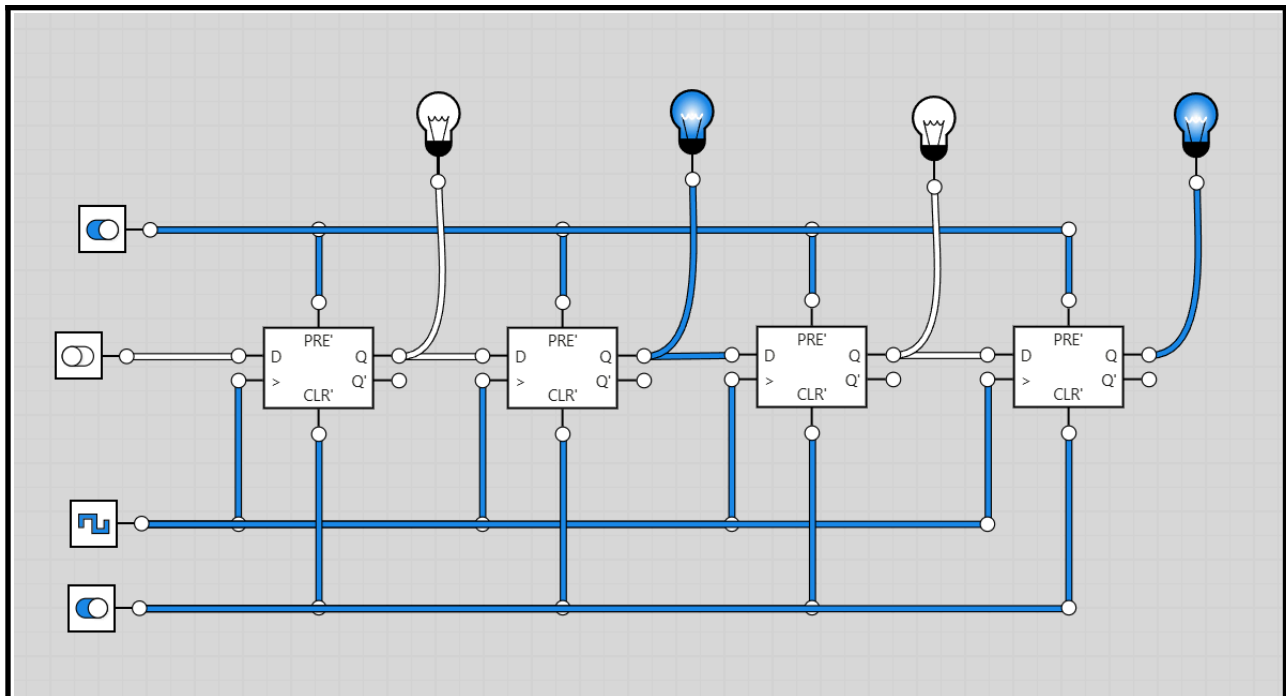
6. Data and Results:

Place here your simulation result and truth table

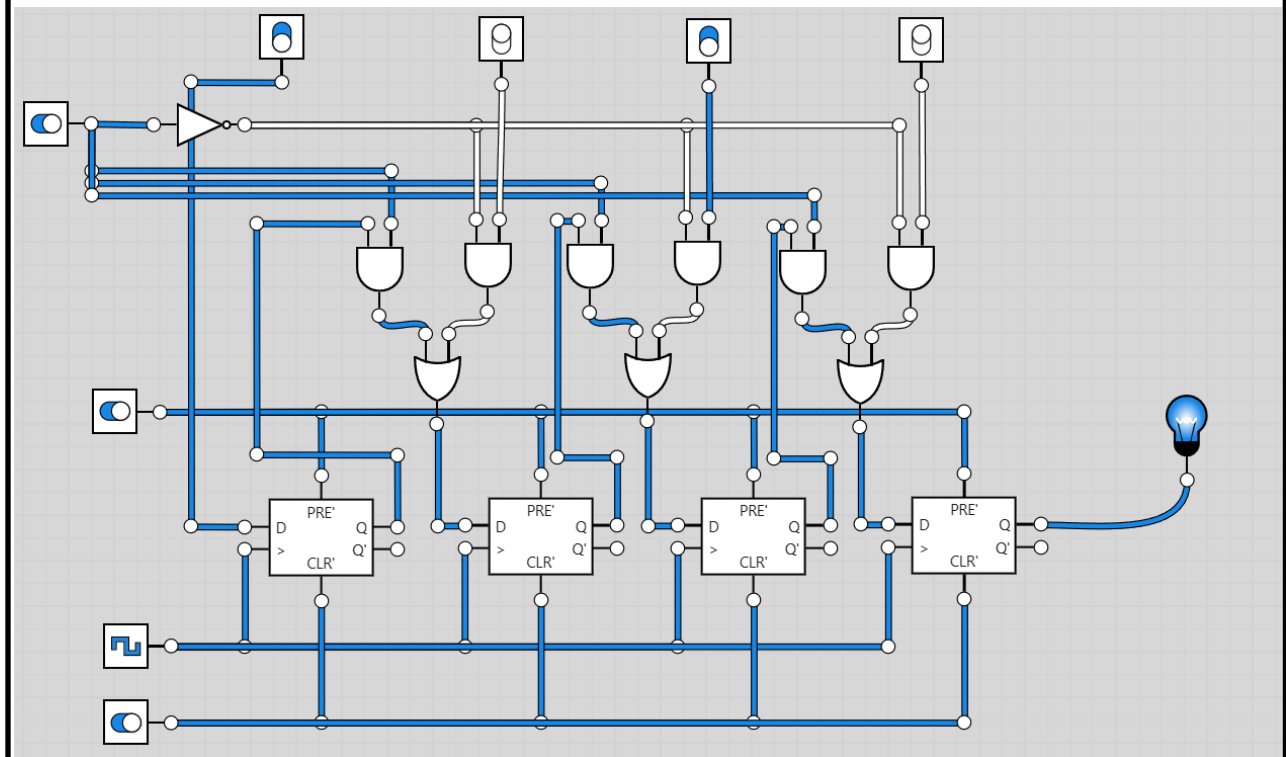
SISO (Logicly)



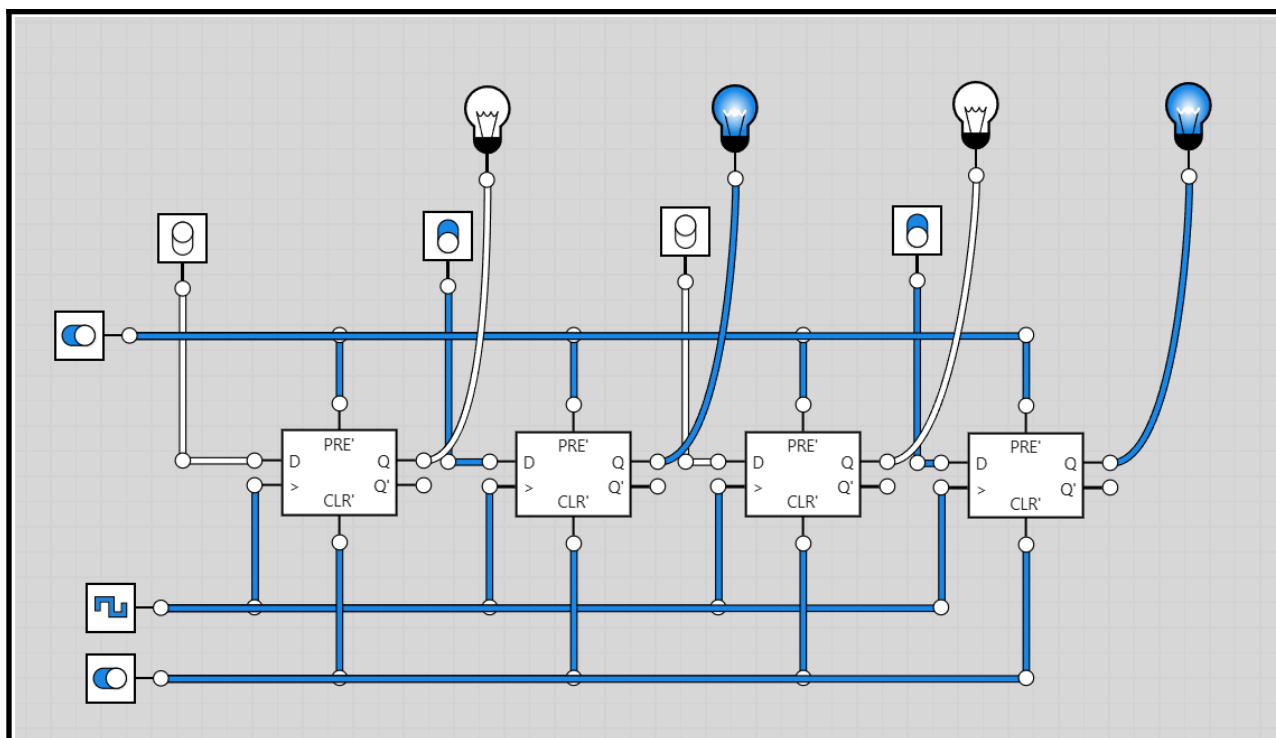
SIPO (Logicly)



PISO (Logicly)



PIPO (Logicly)



7. Final Output:

Screenshot or print screen of Logically application different output simulations (5 points deductions for only one output in every required circuit). Also, upload a video of the simulated circuit naming or talking about the shown input combinations. Failure to comply to each instruction will result to 5 points deductions. Please include the recorded simulation video in this section and write at least one sentence definition or label for each figure or image.

https://drive.google.com/drive/folders/10JfWwcS2A23zg52GnfSA91M_SRPRdfJh?usp=drive_link

8. Conclusion:

Write a one paragraph (at least 10 sentences) Conclusion for this activity.

In this experiment, we successfully implemented and simulated different types of shift registers using Logically, namely SISO, SIPO, PISO, and PIPO. Through this activity, we understood that a shift register is composed of multiple flip-flops used to store and transfer binary data. Each mode of shift register has a unique way of handling input and output data. The SISO register transfers data serially from input to output. The SIPO allows serial input but produces parallel output. On the other hand, PISO accepts parallel input and shifts it out serially. The PIPO handles both input and output in parallel form. During the simulation, we observed how data bits move from one flip-flop to another with each clock pulse. We also created truth tables based on the simulation results. Overall, this experiment enhanced our knowledge of sequential logic circuits and their real-world applications.

9. Questions:

1. Which among the mode of shift register do you think is most used in the industry and other devices? Why?

the most commonly used shift register in the industry is the **Serial Input Parallel Output (SIPO)**. We believe this is because many devices receive data in a serial manner but need to process it in parallel form. SIPO makes it easier to convert serial data into parallel output, which improves speed and efficiency. It also helps reduce the number of input pins and wiring needed in a system. Because of these advantages, it is widely used in microcontrollers, communication systems, and digital displays.

2. Cite different application of register in the industry.

Registers are used in different applications in the industry such as storing temporary data in digital systems and computers (Mano, 2013). They are also used for transferring data between different components of a circuit. Registers are important in communication systems for converting serial data to parallel and vice versa. They are used in LED displays to control multiple outputs using fewer pins. Registers are also applied in digital counters and sequence generators. In addition, they are used in microprocessors for instruction execution and data processing.

10. Assessment (Rubric for Laboratory Performance):

Laboratory Performance Rubric (2)					
Criteria	Ratings			Pts	
Manipulative Skills	3 pts Proficient Student always demonstrate needed skills.	2 pts Acceptable Student occasionally demonstrate needed skills.	1 pts Beginner Student do not demonstrate needed skills.	3 pts	
Experimental Set - up	3 pts Proficient Student are able to set-up the material with minimum supervision.	2 pts Acceptable Student are able to set-up the materials with supervision.	1 pts Beginner Student are unable to set-up the materials.	3 pts	
Process Skills	3 pts Proficient Student always demonstrate targeted process skills.	2 pts Acceptable Student occasionally demonstrate targeted process skills.	1 pts Beginner Student do not demonstrate targeted process skills.	3 pts	
Application of Technical Writing Skills	3 pts Proficient Correct spelling, grammar, and sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	2 pts Acceptable Some minor incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	1 pts Beginner Many incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	3 pts	
Time Management / Conduct of Experiment	3 pts Proficient Student finish ahead of time with complete data and time to revise data.	2 pts Acceptable Student finish on time with incomplete data.	1 pts Beginner Student do not finish on time with incomplete data.	3 pts	
Complete Requirements of Laboratory Report	3 pts Proficient Student submitted a complete requirements set by the teacher.	2 pts Acceptable Student submitted an almost complete Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student submitted a lot of missing requirements in the Laboratory Report according to requirements set by the teacher.	3 pts	
Follow the Correct Format of Laboratory Report	3 pts Proficient Student submitted a correct format of Laboratory Report according to requirements set by the teacher.	2 pts Acceptable Student submitted an almost correct format of Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student did not follow at all the format of Laboratory Report according to requirements set by the teacher.	3 pts	
Ability to do independent work	3 pts Proficient Student do not need to be supervised by the teacher.	2 pts Acceptable Student require occasional supervision by the teacher.	1 pts Beginner Student require supervision by the teacher.	3 pts	
Total Points: 24					